

YIMO Scoring System

Youth International Math Olympiad

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Motivation

YIMO runs two independent divisions. Each division holds two preliminary windows, A and B, on *different* problem sets. The top eight competitors from each window advance to a single divisional Finals. Two competitors of equal ability who sit different windows should earn comparable standing regardless of which problem set they drew. Raw score totals do not deliver this: a window with an easier set inflates its competitors' scores and distorts the ranking.

The system below removes that luck factor. It weights each preliminary problem by how rarely it was solved, then places both windows on a common scale before any cross-window comparison. Finals, a shared proof-based round, are scored on fixed point values and carry double weight in the final standing.

1 Preliminary Scoring

1.1 Problem weights

All preliminary problems are open-ended, so no guessing correction is required. Within a single window-division pool, if a problem is solved by $S > 0$ competitors out of N who took that test, its weight is

$$w = 1 + \ln\left(\frac{N}{S}\right).$$

A competitor's preliminary score is the sum of the weights of every problem they solved. Weights are computed inside each window-division pool, because only those competitors saw that exact problem set.

1.2 Why this weight

- A problem few solve carries a high weight. A problem nearly everyone solves approaches weight 1. Difficulty is measured by the field's actual performance, not by an estimate fixed before the contest.
- The function is continuous and self-calibrating. It produces a clean spread of weights whether the field is 80 competitors or 300, with no buckets or tunable constants.
- A problem solved by nobody contributes no weight rather than an undefined value, so an unused or flawed problem cannot distort scores.

2 Cross-Window Normalization

Because Windows A and B use different problem sets, raw preliminary scores are not directly comparable. Two normalizations place them on a shared scale. Finalists and non-finalists are normalized by different methods, by design, since the two groups are ranked independently and never against each other.

2.1 Finalists: anchor normalization

Within each window, the eight finalists' raw preliminary scores are mapped linearly so the top score becomes 1 and the lowest becomes 0:

$$\text{norm} = \frac{\text{raw} - \text{pool}_{\min}}{\text{pool}_{\max} - \text{pool}_{\min}}.$$

Anchoring both ends forces Window A and Window B finalists onto the identical $[0, 1]$ scale, which is required before their scores can be compared. A normalized preliminary score of 0.85 then means the same thing in either window: 85 percent of the way from the weakest qualifier to the strongest in that window.

2.2 Non-finalists: standardization

Non-finalists form a large pool ranked internally for awards across both windows. Each non-finalist's score is standardized within their own window:

$$z = \frac{\text{raw} - \mu_{\text{window}}}{\sigma_{\text{window}}},$$

where μ and σ are the mean and standard deviation of that window's non-finalist scores. The two windows are then merged and ranked by z . Standardization equalizes the two distributions while preserving margin of victory, information that a simple percentile rank would discard. This assumes both windows were written to comparable difficulty, which the problem-setting process enforces.

3 Finals Scoring

The Finals round is four proof problems, shared by all 16 finalists in a division, graded on partial credit. Solve-rate weighting is *not* applied here. Proofs already encode difficulty through their rubric and partial-credit scale. Applying solve-rate weighting on top would double-count difficulty and is undefined for non-binary scores.

Each problem carries a fixed point value, graduated by intended difficulty:

Problem	P1	P2	P3	P4	Total
Points	5	7	8	10	30

A finalist's raw Finals score is the sum of points earned across the four problems. The 16 raw Finals scores are then anchor-normalized to $[0, 1]$ by the same formula used for finalist preliminaries.

4 Final Standing

4.1 Finalists

A finalist's final score combines the two normalized components:

$$\text{final} = \text{norm}_{\text{prelim}} + 2 \times \text{norm}_{\text{finals}}.$$

Finals carry double weight. The Finals round is harder, shared by all finalists, and free of the cross-window comparability problem, so it should decide the title more than a preliminary that was split across two non-comparable sets. The preliminary still counts: a dominant preliminary performance is not erased, but a strong Finals can overtake it.

4.2 Non-finalists

Non-finalists are ranked by their standardized preliminary score across both windows. Finalists' scores do not affect non-finalist standing.